

Data Warehouse with Medallion Architecture using SQL Server – Naming Conventions

This document outlines the project naming conventions for schemas, tables, views, columns, and other database objects.

1. General Principles:

- **Naming Conventions:** Use snake_case, with lowercase letters and underscores (`\_`) to separate words.
- **Language:** Use English for all names.
- **Avoid Reserved Words:** Do not use SQL reserved words as object names.

2. Table Naming Conventions

2.1 Bronze Rules

- All names must start with the source system name, and table names must match their original names without renaming.
- **<sourcesystem>_<entity>**
 - **<sourcesystem>:** Name of the source system (e.g., crm, erp).
 - **<entity>:** Exact table name from the source system.

- Example: `crm_customer_info` → Customer information from the CRM system.

2.2 Silver Rules

- All names must start with the source system name, and table names must match their original names without renaming.
- **<sourcesystem>_<entity>**
 - **<sourcesystem>**: Name of the source system (e.g., `crm`, `erp`).
 - **<entity>**: Exact table name from the source system.
 - Example: `crm_customer_info` → Customer information from the CRM system.

2.3 Gold Rules

- All names must use meaningful, business-aligned names for tables, starting with the category prefix.
- **<category>_<entity>**
 - **<category>**: Describes the role of the table, such as `dim` (dimension) or `fact` (fact table).
 - **<entity>**: Descriptive name of the table, aligned with the business domain (e.g., `customers`, `products`, `sales`).

- Example:

dim_customers → Dimension table for customer data.

fact_sales → Fact table containing sales transactions.

2.4 Glossary of Category Patterns

Pattern	Meaning	Example(s)
dim_	Dimension Table	dim_customer , dim_product
fact_	Fact Table	fact_sales
agg_	Aggregate Table	agg_customers , agg_sales_monthly

3. Column Naming Conventions

3.1 Surrogate Keys

- All primary keys in dimension tables must use the suffix `_key`.
- `<table_name>_key`
 - `<table_name>`: Refers to the name of the table or entity the key belongs to.
 - `_key`: A suffix indicating that this column is a surrogate key.
 - Example: `customer_key` → Surrogate key in the `dim_customers` table.

3.2 Technical Columns

- All technical columns must start with the prefix `dwh_`, followed by a descriptive name indicating the column's purpose.
- **`dwh_<column_name>`**
 - `dwh`: Prefix exclusively for system-generated metadata.
 - `<column_name>`: Descriptive name indicating the column's purpose.
 - Example: `dwh_load_date` → System-generated column used to store the date when the record was loaded.

4. Stored Procedure

- All stored procedures used for loading data must follow the naming pattern: **`load_<layer>`** .
 - `<layer>`: Represents the layer being loaded, such as bronze, silver, or gold.
 - Example:

`load_bronze` → Stored procedure for loading data into the Bronze layer.

`load_silver` → Stored procedure for loading data into the Silver layer.